



Lesson 2.5 — Completing the Square

Big idea: Completing the square turns any quadratic into a perfect square plus or minus a constant. It's the engine behind both vertex form and the quadratic formula.

Quick review

The move: for $x^2 + bx$, add $(b/2)^2$ to make a perfect square: $(x + b/2)^2$.

Balance: whatever you add inside, subtract back outside (or to the other side of an equation).

Leading coefficient: if $a \neq 1$, factor it out of the x^2 and x terms first.

Used for: converting standard $\rightarrow$ vertex form, deriving the quadratic formula, solving quadratics.

Worked example

Worked example. Solve $x^2 - 6x - 7 = 0$ by completing the square.

$x^2 - 6x = 7$. Add $(-6/2)^2 = 9$ to both sides: $x^2 - 6x + 9 = 16$. $(x - 3)^2 = 16$.

$x - 3 = \pm 4 \Rightarrow x = 7$ or $x = -1$.

Warm-ups

1. What number completes the square: $x^2 + 10x + \underline{\hspace{1cm}}$?
2. What number completes the square: $x^2 - 14x + \underline{\hspace{1cm}}$?
3. Solve $(x - 5)^2 = 36$ for x .

Practice

1. Solve $x^2 + 4x - 12 = 0$ by completing the square.
2. Convert $y = x^2 - 8x + 5$ to vertex form by completing the square.
3. Solve $2x^2 - 8x - 6 = 0$ by completing the square.

Challenge

1. Convert $y = 3x^2 + 12x + 5$ to vertex form, and state the minimum value of the function.
2. Use completing the square on the general quadratic $ax^2 + bx + c = 0$ to derive the quadratic formula. (Show the key algebra steps.)



Answer key — Lesson 2.5

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. 25.
2. 49.
3. $x - 5 = \pm 6 \Rightarrow x = 11$ or $x = -1$.

Practice

1. $x^2 + 4x = 12$; add 4: $(x + 2)^2 = 16$; $x + 2 = \pm 4$; $x = 2, -6$.
2. $x^2 - 8x = (x - 4)^2 - 16$; $y = (x - 4)^2 - 11$.
3. Divide by 2: $x^2 - 4x - 3 = 0 \rightarrow x^2 - 4x = 3 \rightarrow (x - 2)^2 = 7 \rightarrow x = 2 \pm \sqrt{7}$.

Challenge

1. Factor 3: $3(x^2 + 4x) + 5 = 3[(x + 2)^2 - 4] + 5 = 3(x + 2)^2 - 7$. Minimum = -7 (at $x = -2$).
2. Divide by a ; $x^2 + (b/a)x = -c/a$; complete the square: $(x + b/(2a))^2 = b^2/(4a^2) - c/a = (b^2 - 4ac)/(4a^2)$; take square root and isolate $x \rightarrow x = (-b \pm \sqrt{b^2 - 4ac})/(2a)$. ✓